BANGALORE NORTH UNIVERSITY

KOLAR, KARNATAKA-563103

INTERNSHIP REPORT ON

"TOY STORE"

in the domain

"FULL STACK WEBDEVLOPMENT"

SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE OF BACHELOR OF COMPUTER APPLICATIONS

BY

MUTHESH K (U19KU22S0111)

Under the guidance of

MS.NAMRATA

Sr. Assistant Professor

Department of BCA

&

PRINSTON SMART ENGINEERS

New Horizon College, Marathalli

Bengaluru-560103

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STUDENT DECLARATION

I MUTHESH K NUMBER: U19KU22S0111, hereby declare that this internship report entitled “TOY STORE” under the domain “FULL STACK WEBDEVLOPMENT” is prepared by me during the internship period from 08/02/25 to 08/03/25 at PRINSTON SMART ENGINEERS under the supervision and guidance of MS.NAMRATA, Assistant Professor of BCA Department, New Horizon College, Marathahalli, Bengaluru.

Date:

Place:

Signature

Name : MUTHESH K

Reg No: U19KU22S0111

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Chapter I
INTRODUCTION

The **Toyfinity Toy Store Web Application** is a full-stack e-commerce platform developed to offer users a seamless and engaging experience for browsing and purchasing toys online. Built with **React.js** on the frontend, **Node.js/Express** on the backend, and **MongoDB** as the database, the application replicates essential features of a modern online shopping system in a compact and educational format.

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2. Purpose of the Project:

The primary purpose of this project is to:

- Provide a **user-friendly interface** for toy browsing and buying.
- Simulate a complete **e-commerce flow**, from product display to checkout.
- Serve as a **learning and demonstration tool** for full-stack web development using the MERN (MongoDB, Express, React, Node.js) stack.

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3. Key Features:

- **Home Page** with animated greeting and background.
- **Products Page** with search functionality and "Add to Cart" buttons.
- **Cart Page** with quantity updates, item removal, and total calculation.
- **Checkout Page** with shipping address and multiple payment options (COD, UPI, Card).
- **Order History** page showing past orders for logged-in users.
- **Login/Register** system with localStorage-based authentication.
- **CartContext** using React's Context API for global cart state.
- **Responsive Design** and smooth UI with animations and hover effects.

- **Welcome message** and dynamic Navbar showing cart count and user info.
- Option to **logout or delete account** from the Navbar.

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4. Benefits:

- Enhances understanding of real-world web development workflows.
- Offers hands-on experience with frontend and backend integration.
- Helps in practicing **React state management** and API handling.
- Can serve as a **portfolio project** or **academic assignment**.
- Modular and extendable – easy to integrate more features (e.g., admin panel, Stripe, etc.).

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5. About the Project:

Toyfinity is a fictional online toy shop built as a full-stack educational project. It emphasizes simplicity in design and clarity in structure. The backend handles API logic and data persistence, while the frontend ensures a responsive, attractive user interface. The application follows a clean separation of concerns and is optimized for local and cloud deployment.

Chapter 2
REQUIREMENT SPECIFICATION

2 Software Requirements:

Frontend:

- **Framework: React.js**
- **Dependencies:**
  - **React Router**
  - **Context API**
  - **Axios**
  - **Framer Motion (optional for animation)**
- **Build Tool: react-scripts (via Create React App)**
- **Styling: Plain CSS**
- **Browser Support: Latest versions of Chrome, Firefox, Edge, Safari**

Backend:

- **Runtime Environment: Node.js (v14 or above)**
- **Framework: Express.js**
- **Database Driver: Mongoose (for MongoDB)**
- **Other Libraries:**
  - **cors, dotenv, nodemon (for development)**
  - **body-parser (if needed)**

Database:

- **Database: MongoDB (local or cloud e.g., MongoDB Atlas)**
- **Schema Management: Mongoose**

Development Tools:

- **Code Editor: Visual Studio Code**
- **Version Control: Git**
- **Package Manager: npm**
- **Browser DevTools: Chrome Developer Tools**

Chapter 3

SOFTWARE DESIGN

1. Frontend Design (Client-Side)

Technology:

- **React.js** (JavaScript)
- **React Router** for navigation
- **Context API** for global cart state
- **CSS** for styling and animations

Key Features:

- **Home Page** with background image, animation, and "Shop Now" button.
- **Products Page**: Displays product cards, search/filter functionality.
- **Cart Page**: Lists selected items, allows quantity changes and deletion.
- **Checkout Page**: Collects shipping address and payment method (COD/UPI/Card).
- **Order History Page**: Displays user's past orders.
- **Login/Register Pages**: Basic user authentication using localStorage.
- **Navbar/Footer**: Dynamic links, user greeting, cart count, and logout/delete account options.

---

2. Backend Design (Server-Side)

Technology:

- **Node.js + Express.js**

Key Responsibilities:

- Handles API requests (CRUD operations).

- Serves product and order data.
- Processes new order submissions.
- Performs basic user registration/login with local storage simulation (no real session management).

API Endpoints:

- GET /api/products – Fetch/search all products.
- POST /api/orders – Save a new order.
- GET /api/orders – Retrieve order history.
- *(Optional)* POST /api/register, POST /api/login – Simulated local authentication.

---

3. Database Design

Technology:

- **MongoDB** (with **Mongoose** ODM)

Collections:

- **Users**
  - name, email, password
- **Products**
  - name, description, price, image, category
- **Orders**
  - items, total, shippingAddress, paymentMethod, date

Relations:

- Orders are linked to users by email or ID.
- Products are referenced inside order items.

Summary:

This architecture separates concerns cleanly:

- The **frontend** handles UI/UX and user interactions.
- The **backend** manages data and logic.
- The **database** stores structured data for persistence.

IMPLEMENTATION :

Frontend codes:

HTML File - index.html

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" href="%PUBLIC_URL%/favicon.ico" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <meta name="theme-color" content="#254dff" />
    <meta name="description" content="Toyfinity - Your Favorite Toy Store" />
    <title>Toyfinity</title>
  </head>
  <body>
    <noscript>You need to enable JavaScript to run this app.</noscript>
    <div id="root"></div>
  </body>
</html>
```

JavaScript File - CartContext.js

```
import React, { createContext, useState, useEffect } from 'react';

export const CartContext = createContext();

export function CartProvider({ children }) {
  const [cartItems, setCartItems] = useState(() => {
    const stored = localStorage.getItem('cart');
    return stored ? JSON.parse(stored) : [];
  });

  useEffect(() => {
    localStorage.setItem('cart', JSON.stringify(cartItems));
  }, [cartItems]);

  const addToCart = (product) => {
```

```
    setCartItems((prev) => {
      const exists = prev.find((item) => item._id === product._id);
      if (exists) {
        return prev.map((item) =>
          item._id === product._id ? { ...item, quantity: item.quantity + 1 } : item
        );
      }
      return [...prev, { ...product, quantity: 1 }];
    });
  };

  const removeFromCart = (productId) => {
    setCartItems((prev) => prev.filter((item) => item._id !== productId));
  };

  const updateQuantity = (productId, quantity) => {
    setCartItems((prev) =>
      prev.map((item) =>
        item._id === productId ? { ...item, quantity } : item
      )
    );
  };

  return (
    <CartContext.Provider value={{ cartItems, addToCart, removeFromCart,
updateQuantity }}>
      {children}
    </CartContext.Provider>
  );
}
```

JavaScript File - Footer.js

```
import React from 'react';
import './Footer.css';

function Footer() {
  return (
    <footer className="footer">
      <div className="footer-content">
        <p> 📍 Toyfinity Store, #18, Ruby Layout, Bengaluru, India</p>
```

```
      <p>📞 Contact: +91-8867082099 | ✉️ Email:
muthesh2004@gmail.com</p>
      <p>© {new Date().getFullYear()} Toyfinity. All rights reserved.</p>
    </div>
   </footer>
  );
}

export default Footer;
```

CSS File - Navbar.css

```
.navbar {
  display: flex;
  justify-content: space-between;
  align-items: center;
  background-color: #254dff;
  padding: 10px 30px;
  color: white;
  box-shadow: 0 2px 8px rgba(0, 0, 0, 0.15);
}

.logo {
  display: flex;
  align-items: center;
  font-size: 1.8rem;
  font-weight: bold;
  color: white;
  text-decoration: none;
  gap: 6px;
  transition: transform 0.3s ease;
}

.logo:hover {
  transform: scale(1.05);
}

.logo-img {
  height: 40px;
  width: 40px;
  object-fit: contain;
```

```
}

.nav-links {
  list-style: none;
  display: flex;
  align-items: center;
  gap: 20px;
}

.nav-links a {
  color: white;
  text-decoration: none;
  font-weight: 500;
  transition: color 0.2s ease;
}

.nav-links a:hover {
  color: #ffcc00;
}

.logout-btn {
  background-color: white;
  color: #254dff;
  border: none;
  padding: 6px 12px;
  cursor: pointer;
  border-radius: 5px;
  font-weight: bold;
  transition: background 0.3s, transform 0.2s ease;
}

.logout-btn:hover {
  background-color: #e0e0e0;
  transform: scale(1.05);
}

.welcome-box {
  padding: 6px 12px;
  background: rgba(255, 255, 255, 0.1);
  border: 1px solid #ffcc00;
  border-radius: 10px;
  animation: fadeInUser 0.5s ease;
```

```
  color: #ffcc00;
  font-weight: bold;
  backdrop-filter: blur(4px);
  box-shadow: 0 2px 6px rgba(255, 204, 0, 0.4);
}

.welcome-box span {
  color: #fff;
}

@keyframes fadeInUser {
  from {
    opacity: 0;
    transform: translateY(-5px);
  }
  to {
    opacity: 1;
    transform: translateY(0);
  }
}
```

JavaScript File - Navbar.jsx

```
import React, { useContext, useEffect, useState } from 'react';
import { Link, useNavigate } from 'react-router-dom';
import './Navbar.css';
import { CartContext } from '../context/CartContext';

function Navbar() {
  const { cartItems } = useContext(CartContext);
  const [user, setUser] = useState(null);
  const navigate = useNavigate();

  useEffect(() => {
    const storedUser = localStorage.getItem('currentUser');
    if (storedUser) {
      setUser(JSON.parse(storedUser));
    }
  }, []);

  const handleLogout = () => {
```

```
    const action = window.prompt("Type 'delete' to delete your account or
'logout' to logout:");

    if (action === 'delete') {
      const currentUser = JSON.parse(localStorage.getItem('currentUser'));
      if (currentUser) {
        const users = JSON.parse(localStorage.getItem('users') || '[]');
        const updatedUsers = users.filter(u => u.email !== currentUser.email);
        localStorage.setItem('users', JSON.stringify(updatedUsers));
      }
      localStorage.removeItem('currentUser');
      alert('Account deleted.');
      setUser(null);
      navigate('/register');
    } else if (action === 'logout') {
      localStorage.removeItem('currentUser');
      alert('Logged out.');
      setUser(null);
      navigate('/login');
    } else {
      alert("Invalid input. Please type 'delete' or 'logout'.");
    }
  };

  const cartCount = cartItems.reduce((sum, item) => sum + item.quantity, 0);

  return (
    <nav className="navbar">
      <Link to="/" className="logo">
        <img src="https://encrypted-
tbn0.gstatic.com/images?q=tbn:ANd9GcTSUZiUiKZj3XBKNMbOGFToMj4DANw
z9gDPuw&s" alt="Logo" className="logo-img" />
        Toyfinity
      </Link>
      <ul className="nav-links">
        <li><Link to="/">Home</Link></li>
        <li><Link to="/products">Products</Link></li>
        <li><Link to="/cart">Cart ({cartCount})</Link></li>
        {user ? (
          <>
            <li><Link to="/orders">Order History</Link></li>
            <li className="welcome-box">Welcome, <span>{user.name ||
```

```
user.email}</span>!</li>
          <li><button className="logout-btn"
onClick={handleLogout}>Logout</button></li>
        </>
       ) : (
        <>
          <li><Link to="/login">Login</Link></li>
          <li><Link to="/register">Register</Link></li>
        </>
       )}
     </ul>
   </nav>
  );
}

export default Navbar;
```

CSS File - Footer.css

```
.footer {
  background-color: #254dff;
  color: white;
  text-align: center;
  padding: 25px 20px;
  font-size: 0.95rem;
  box-shadow: 0 -2px 10px rgba(0, 0, 0, 0.1);
  margin: 0; /* Removed margin-top to avoid white space */
}

.footer-content {
  max-width: 900px;
  margin: auto;
}

.footer-content p {
  margin: 6px 0;
  line-height: 1.6;
}

.footer-content a {
  color: #ffeb3b;
```

```
  text-decoration: none;
  font-weight: 500;
}

.footer-content a:hover {
  text-decoration: underline;
}

/* Icon style if using emojis or Font Awesome */
.footer-content .icon {
  margin-right: 6px;
}
```

CSS File - App.css

```
body {
  margin: 0;
  font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
  background-color: #f5f5f5;
  color: #333;
}

.app-content {
  min-height: 80vh;
  padding-top: 20px;
  padding-bottom: 20px;
}

a {
  text-decoration: none;
  color: inherit;
}
```

CSS File - index.css

```
body {
  margin: 0;
  padding: 0;
  font-family: 'Segoe UI', sans-serif;
  background-color: #f5f5f5;
}
```

```
body {
  font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
  background-color: #f6f6f6;
  color: #333;
  line-height: 1.6;
}

a {
  text-decoration: none;
  color: inherit;
}

button {
  font-family: inherit;
}
```

JavaScript File - App.js

```
import React from 'react';
import { BrowserRouter as Router, Routes, Route } from 'react-router-dom';
import { CartProvider } from './context/CartContext';

import Home from './pages/Home';
import Products from './pages/Products';
import Cart from './pages/Cart';
import Checkout from './pages/Checkout';
import Confirmation from './pages/Confirmation';
import Orders from './pages/Orders';
import Login from './pages/Login';
import Register from './pages/Register';

import Navbar from './components/Navbar';
import Footer from './components/Footer';

function App() {
  return (
    <CartProvider>
      <Router>
        <Navbar />
        <div style={{ minHeight: '80vh' }}>
```

```
        <Routes>
          <Route path="/" element={<Home />} />
          <Route path="/products" element={<Products />} />
          <Route path="/cart" element={<Cart />} />
          <Route path="/checkout" element={<Checkout />} />
          <Route path="/confirmation" element={<Confirmation />} />
          <Route path="/orders" element={<Orders />} />
          <Route path="/login" element={<Login />} />
          <Route path="/register" element={<Register />} />
        </Routes>
      </div>
      <Footer />
    </Router>
  </CartProvider>
 );
}

export default App;
```

JavaScript File - index.js

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';
import './index.css';

const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(<App />);
```

CSS File - Orders.css

```
.orders-container {
  max-width: 900px;
  margin: auto;
  padding: 30px 20px;
}

.orders-container h2 {
  text-align: center;
  margin-bottom: 20px;
}
```

```
.order-card {
  background: #fff;
  padding: 20px;
  border-radius: 8px;
  margin-bottom: 20px;
  box-shadow: 0 2px 6px rgba(0,0,0,0.1);
}

.order-card ul {
  padding-left: 20px;
}
```

CSS File - Home.css

```
/* Reset page spacing */
body, html {
  margin: 0;
  padding: 0;
  overflow-x: hidden;
  height: 100%;
}

/* Hero Section */
.home-hero {
  position: relative;
  min-height: 100vh;
  background-image: url('https://thumbs.dreamstime.com/b/group-baby-toys-
wooden-background-copy-space-top-vie-brown-view-97584351.jpg');
  background-size: cover;
  background-position: center;
  display: flex;
  align-items: center;
  justify-content: center;
  color: white;
  text-align: center;
  overflow: hidden;
}

/* Dark overlay */
.overlay {
```

```
  position: absolute;
  width: 100%;
  height: 100%;
  background-color: rgba(0, 0, 0, 0.55);
  top: 0;
  left: 0;
  z-index: 1;
}

/* Box with animation */
.animated-box {
  position: relative;
  z-index: 2;
  background-color: rgba(255, 255, 255, 0.1);
  border: 2px solid rgba(255, 255, 255, 0.3);
  backdrop-filter: blur(8px);
  border-radius: 16px;
  padding: 40px 30px;
  animation: floatUp 1.5s ease-in-out;
  box-shadow: 0 8px 32px rgba(0, 0, 0, 0.25);
  max-width: 600px;
}

.animated-box h1 {
  font-size: 3rem;
  margin-bottom: 15px;
  color: #fff3a3;
}

.user-greet {
  font-size: 1.3rem;
  color: #ffcc00;
  margin-bottom: 10px;
}

.user-greet span {
  font-weight: bold;
  color: #fff;
  text-shadow: 0 0 5px #ffcc00;
}

.description {
```

```
  font-size: 1.1rem;
  margin-bottom: 30px;
  color: #e8f8f9;
}

.shop-btn {
  padding: 14px 28px;
  font-size: 18px;
  background-color: #254dff;
  border: none;
  border-radius: 8px;
  color: white;
  cursor: pointer;
  transition: background 0.3s ease, transform 0.2s ease;
}

.shop-btn:hover {
  background-color: #1f3fd7;
  transform: scale(1.05);
}

@keyframes floatUp {
  from {
    opacity: 0;
    transform: translateY(50px);
  }
  to {
    opacity: 1;
    transform: translateY(0);
  }
}
```

JavaScript File - Home.jsx

```
import React, { useEffect, useState } from 'react';
import './Home.css';
import { Link } from 'react-router-dom';

function Home() {
  const [user, setUser] = useState(null);
```

```
  useEffect(() => {
    const storedUser = localStorage.getItem('currentUser');
    if (storedUser) {
      setUser(JSON.parse(storedUser));
    }
  }, []);

  return (
    <div className="home-hero">
      <div className="overlay"></div>
      <div className="animated-box">
        <h1>Welcome to our ToyStore!</h1>
        {user && <p className="user-greet">Hello, <span>{user.name ||
user.email}</span> !</p>}
        <p className="description">Discover fun, joy, and learning in every Toy
!!</p>
        <Link to="/products">
          <button className="shop-btn">Shop Now</button>
        </Link>
      </div>
    </div>
  );
}

export default Home;
```

CSS File - Register.css

```
.register-container {
  width: 100%;
  max-width: 400px;
  margin: 60px auto;
  padding: 30px;
  background-color: #ffffff;
  border-radius: 10px;
  box-shadow: 0 4px 12px rgba(0, 0, 0, 0.1);
}

.register-container h2 {
  text-align: center;
  margin-bottom: 20px;
```

```
  color: #254dff;
}

.register-container input {
  display: block;
  width: 100%;
  padding: 12px;
  margin: 10px 0;
  font-size: 16px;
  border: 1px solid #ccc;
  border-radius: 6px;
  box-sizing: border-box;
}

.register-container button {
  width: 100%;
  background-color: #254dff;
  color: white;
  padding: 12px;
  border: none;
  border-radius: 6px;
  font-size: 16px;
  cursor: pointer;
  margin-top: 10px;
  transition: background-color 0.3s ease;
}

.register-container button:hover {
  background-color: #1930b8;
}

.register-container p {
  text-align: center;
  margin-top: 15px;
  font-size: 14px;
}

.register-container a {
  color: #254dff;
  text-decoration: none;
  font-weight: bold;
```

```
}
```

CSS File - Login.css

```
.auth-container {
  width: 100%;
  max-width: 400px;
  margin: 60px auto;
  padding: 30px;
  background-color: #ffffff;
  border-radius: 10px;
  box-shadow: 0 4px 12px rgba(0, 0, 0, 0.1);
}

.auth-container h2 {
  text-align: center;
  margin-bottom: 20px;
  color: #254dff;
}

.auth-container input {
  display: block;
  width: 100%;
  padding: 12px;
  margin: 10px 0;
  font-size: 16px;
  border: 1px solid #ccc;
  border-radius: 6px;
  box-sizing: border-box;
}

.auth-container button {
  width: 100%;
  background-color: #254dff;
  color: white;
  padding: 12px;
  border: none;
  border-radius: 6px;
  font-size: 16px;
  cursor: pointer;
  margin-top: 10px;
```

```
  transition: background-color 0.3s ease;
}

.auth-container button:hover {
  background-color: #1930b8;
}

.auth-container p {
  text-align: center;
  margin-top: 15px;
  font-size: 14px;
}

.auth-container a {
  color: #254dff;
  text-decoration: none;
  font-weight: bold;
}
```

JavaScript File - Login.js

```
import React, { useState } from 'react';
import { useNavigate } from 'react-router-dom';
import './Login.css';

function Login() {
  const [email, setEmail] = useState('');
  const [password, setPassword] = useState('');
  const navigate = useNavigate();

  const handleLogin = () => {
    const users = JSON.parse(localStorage.getItem('users') || '[]');
    const user = users.find((u) => u.email === email && u.password ===
password);
    if (user) {
      localStorage.setItem('currentUser', JSON.stringify(user));
      navigate('/');
    } else {
      alert('Invalid credentials');
    }
  };
```

```
  return (
    <div className="auth-container">
      <h2>Login</h2>
      <form onSubmit={(e) => { e.preventDefault(); handleLogin(); }}>
        <input
          type="email"
          placeholder="Email address"
          value={email}
          required
          onChange={(e) => setEmail(e.target.value)}
        />
        <input
          type="password"
          placeholder="Enter your Password"
          value={password}
          required
          onChange={(e) => setPassword(e.target.value)}
        />
        <button type="submit">Login</button>
      </form>
      <p>
        Don’t have an account? <a href="/register">Register here</a>
      </p>
    </div>
  );
}

export default Login;
```

JavaScript File - Register.js

```
import React, { useState } from 'react';
import { useNavigate } from 'react-router-dom';
import './Register.css';

function Register() {
  const [name, setName] = useState('');
  const [email, setEmail] = useState('');
  const [password, setPassword] = useState('');
  const navigate = useNavigate();
```

```
const handleRegister = () => {
  const users = JSON.parse(localStorage.getItem('users') || '[]');
  if (users.find((u) => u.email === email)) {
    alert('User already exists');
    return;
  }

  const newUser = { name, email, password };
  users.push(newUser);
  localStorage.setItem('users', JSON.stringify(users));
  localStorage.setItem('currentUser', JSON.stringify(newUser));
  navigate('/');
};

return (
  <div className="auth-container">
    <h2>Register</h2>
    <form onSubmit={(e) => { e.preventDefault(); handleRegister(); }}>
      <input
        type="text"
        placeholder="Full Name"
        value={name}
        required
        onChange={(e) => setName(e.target.value)}
      />
      <input
        type="email"
        placeholder="Email address"
        value={email}
        required
        onChange={(e) => setEmail(e.target.value)}
      />
      <input
        type="password"
        placeholder="Enter your Password"
        value={password}
        required
        onChange={(e) => setPassword(e.target.value)}
      />
      <button type="submit">Register</button>
    </form>
```

```
    <p>
      Already have an account? <a href="/login">Login here</a>
    </p>
   </div>
  );
}

export default Register;
```

JavaScript File - Products.jsx

```
import React, { useEffect, useState, useContext } from 'react';
import axios from 'axios';
import './Products.css';
import { CartContext } from '../context/CartContext';

function Products() {
  const [products, setProducts] = useState([]);
  const [search, setSearch] = useState('');
  const [feedback, setFeedback] = useState('');
  const { addToCart } = useContext(CartContext);

  useEffect(() => {
    const fetchProducts = async () => {
      try {
        const query = search ? `?search=${search}` : '';
        const res = await axios.get(`http://localhost:5000/api/products${query}`);
        setProducts(res.data);
      } catch (err) {
        console.error('Failed to fetch products:', err);
      }
    };

    fetchProducts();
  }, [search]);

  const handleAddToCart = (product) => {
    addToCart(product);
    setFeedback(` ✅ "${product.name}" added to cart!`);
    setTimeout(() => setFeedback(''), 2000);
  };
```

```
  return (
    <div className="products-container">
      <h2>Our Products</h2>

      <div className="filter-bar">
        <input
          type="text"
          placeholder="Search toys..."
          value={search}
          onChange={(e) => setSearch(e.target.value)}
        />
      </div>

      {feedback && <div className="feedback-message">{feedback}</div>}

      <div className="product-list">
        {products.map((p) => (
          <div key={p._id} className="product-card">
            <img src={p.image} alt={p.name} />
            <h3>{p.name}</h3>
            <p>₹{p.price.toFixed(2)}</p>
            <button onClick={() => handleAddToCart(p)}>Add to Cart</button>
          </div>
        ))}
      </div>
    </div>
  );
}

export default Products;
```

JavaScript File - Cart.jsx

```
import React, { useContext, useEffect } from 'react';
import './Cart.css';
import { CartContext } from '../context/CartContext';
import { Link } from 'react-router-dom';

function Cart() {
  const { cartItems, updateQuantity, removeFromCart } =
```

```
useContext(CartContext);

  useEffect(() => {
    localStorage.setItem('cart', JSON.stringify(cartItems));
  }, [cartItems]);

  const incrementQty = (id) => {
    const item = cartItems.find((item) => item._id === id);
    updateQuantity(id, item.quantity + 1);
  };

  const decrementQty = (id) => {
    const item = cartItems.find((item) => item._id === id);
    if (item.quantity > 1) {
      updateQuantity(id, item.quantity - 1);
    }
  };

  const total = cartItems.reduce((sum, item) => sum + item.price *
item.quantity, 0);

  return (
    <div className="cart-container">
      <h2>Your Cart</h2>
      {cartItems.length === 0 ? (
        <div className="empty-cart">
          <p>Your cart is empty.</p>
          <Link to="/products">
            <button className="shop-now-btn">Shop Now</button>
          </Link>
        </div>
      ) : (
        <>
          {cartItems.map(item => (
            <div key={item._id} className="cart-item">
              <img src={item.image} alt={item.name} />
              <div className="details">
                <h4>{item.name}</h4>
                <p>₹{item.price.toFixed(2)}</p>
                <div className="qty-controls">
                  <button onClick={() => decrementQty(item._id)}>-</button>
                  <span>{item.quantity}</span>
```

```
              <button onClick={() => incrementQty(item._id)}>+</button>
            </div>
            <button className="remove-btn" onClick={() =>
removeFromCart(item._id)}>Remove</button>
          </div>
         </div>
        ))}
        <div className="cart-total">
         <h3>Total: ₹{total.toFixed(2)}</h3>
         <Link to="/checkout">
          <button className="checkout-btn">Proceed to Checkout</button>
         </Link>
        </div>
       </>
     )}
   </div>
 );
}

export default Cart;
```

JavaScript File - Orders.js

```
import React, { useEffect, useState } from 'react';
import './Orders.css';

function Orders() {
 const [orders, setOrders] = useState([]);

 useEffect(() => {
  const stored = localStorage.getItem('orders');
  if (stored) {
   setOrders(JSON.parse(stored));
  }
 }, []);

 return (
  <div className="orders-container">
   <h2>Your Orders</h2>
   {orders.length === 0 ? (
    <p>You have no orders yet.</p>
```

```
    ) : (
      orders.map((order, index) => (
        <div key={index} className="order-card">
          <p><strong>Date:</strong> {new
Date(order.date).toLocaleString()}</p>
          <p><strong>Payment:</strong> {order.paymentMethod}</p>
          <p><strong>Total:</strong> ₹{order.total.toFixed(2)}</p>
          <ul>
            {order.items.map((item, idx) => (
              <li key={idx}>{item.name} × {item.quantity}</li>
            ))}
          </ul>
        </div>
      ))
    )}
  </div>
 );
}

export default Orders;
```

CSS File - Products.css

```
.products-container {
  max-width: 1200px;
  margin: auto;
  padding: 30px 20px;
}

.products-container h2 {
  text-align: center;
  margin-bottom: 20px;
}

.filter-bar {
  display: flex;
  justify-content: center;
  margin-bottom: 25px;
}

.filter-bar input {
```

```
  padding: 10px;
  font-size: 16px;
  width: 300px;
  border-radius: 6px;
  border: 1px solid #ccc;
}

.feedback-message {
  background-color: #d1ffd1;
  color: #006600;
  padding: 10px 20px;
  text-align: center;
  margin-bottom: 15px;
  border-radius: 6px;
  animation: slideDown 0.5s ease;
}

.product-list {
  display: grid;
  grid-template-columns: repeat(auto-fill, minmax(220px, 1fr));
  gap: 20px;
}

.product-card {
  background: white;
  border-radius: 8px;
  padding: 15px;
  box-shadow: 0 2px 5px rgba(0, 0, 0, 0.1);
  text-align: center;
  transition: transform 0.3s ease;
}

.product-card:hover {
  transform: translateY(-5px);
}

.product-card img {
  width: 100%;
  height: 160px;
  object-fit: cover;
  border-radius: 6px;
}
```

```
.product-card h3 {
  margin: 10px 0 5px;
}

.product-card p {
  font-weight: bold;
}

.product-card button {
  margin-top: 10px;
  padding: 8px 16px;
  background-color: #254dff;
  color: white;
  border: none;
  border-radius: 4px;
  cursor: pointer;
}

.product-card button:hover {
  background-color: #1e40ff;
}

/* Animation */
@keyframes slideDown {
  from {
    transform: translateY(-15px);
    opacity: 0;
  }
  to {
    transform: translateY(0);
    opacity: 1;
  }
}
```

CSS File - Confirmation.css

```
.confirmation-container {
  max-width: 600px;
  margin: 100px auto;
  text-align: center;
```

```
  background: #fff;
  padding: 40px;
  border-radius: 10px;
  box-shadow: 0 2px 10px rgba(0,0,0,0.1);
}

.view-orders-btn {
  margin-top: 20px;
  padding: 10px 20px;
  background-color: #254dff;
  color: white;
  border: none;
  border-radius: 6px;
  cursor: pointer;
  font-size: 1rem;
}

.view-orders-btn:hover {
  background-color: #1e40ff;
}
```

JavaScript File - Confirmation.js

```
import React from 'react';
import { Link } from 'react-router-dom';
import './Confirmation.css';

function Confirmation() {
  return (
    <div className="confirmation-container">
      <h2>🎉 Order Confirmed!</h2>
      <p>Thank you for shopping with Toyfinity.</p>
      <Link to="/orders">
        <button className="view-orders-btn">View Orders</button>
      </Link>
    </div>
  );
}

export default Confirmation;
```

CSS File - Checkout.css

```
.checkout-container {
  max-width: 1000px;
  margin: auto;
  padding: 30px 20px;
}

.checkout-container h2 {
  text-align: center;
  margin-bottom: 25px;
}

.order-summary {
  background: #fff;
  padding: 20px;
  margin-bottom: 30px;
  border-radius: 8px;
  box-shadow: 0 2px 6px rgba(0,0,0,0.1);
}

.order-summary ul {
  padding-left: 20px;
}

.shipping-form {
  background: #fff;
  padding: 20px;
  border-radius: 8px;
  box-shadow: 0 2px 6px rgba(0,0,0,0.1);
}

.shipping-form textarea,
.shipping-form input,
.shipping-form select {
  width: 100%;
  padding: 10px;
  margin-top: 10px;
  border-radius: 6px;
  border: 1px solid #ccc;
  font-size: 1rem;
  margin-bottom: 15px;
```

```
}

.card-inputs {
  display: flex;
  flex-direction: column;
  gap: 12px;
}

.shipping-form button {
  background-color: #254dff;
  color: white;
  border: none;
  padding: 10px 16px;
  font-size: 1rem;
  border-radius: 6px;
  cursor: pointer;
  width: 100%;
  margin-top: 15px;
}

.shipping-form button:hover {
  background-color: #1e40ff;
}
```

JavaScript File - Checkout.js

```
import React, { useEffect, useState } from 'react';
import { useNavigate } from 'react-router-dom';
import './Checkout.css';

function Checkout() {
  const [cart, setCart] = useState([]);
  const [address, setAddress] = useState('');
  const [paymentMethod, setPaymentMethod] = useState('COD');
  const [upiId, setUpiId] = useState('');
  const [cardDetails, setCardDetails] = useState({ number: '', expiry: '', cvv: '' });

  const navigate = useNavigate();

  useEffect(() => {
    const stored = localStorage.getItem('cart');
```

```
  if (stored) setCart(JSON.parse(stored));
}, []);

const total = cart.reduce((sum, item) => sum + item.price * item.quantity, 0);

const handlePlaceOrder = () => {
  if (!address.trim()) {
    alert('Please enter a shipping address.');
    return;
  }

  if (paymentMethod === 'UPI' && !upiId.trim()) {
    alert('Please enter your UPI ID.');
    return;
  }

  if (paymentMethod === 'Card') {
    const { number, expiry, cvv } = cardDetails;
    if (!number || !expiry || !cvv) {
      alert('Please fill in all card details.');
      return;
    }
  }

  const order = {
    items: cart,
    total,
    shippingAddress: address,
    paymentMethod,
    paymentDetails:
      paymentMethod === 'UPI' ? { upiId } :
      paymentMethod === 'Card' ? { ...cardDetails } : null,
    date: new Date().toISOString(),
  };

  const orders = JSON.parse(localStorage.getItem('orders') || '[]');
  orders.push(order);
  localStorage.setItem('orders', JSON.stringify(orders));
  localStorage.removeItem('cart');

  navigate('/confirmation');
};
```

```
  return (
    <div className="checkout-container">
      <h2>Checkout</h2>
      {cart.length === 0 ? (
        <p>Your cart is empty.</p>
      ) : (
        <>
          <div className="order-summary">
            <h3>Order Summary</h3>
            <ul>
              {cart.map(item => (
                <li key={item._id}>
                  {item.name} x {item.quantity} = ₹{(item.price *
item.quantity).toFixed(2)}
                </li>
              ))}
            </ul>
            <p><strong>Total: ₹{total.toFixed(2)}</strong></p>
          </div>

          <div className="shipping-form">
            <h3>Shipping Address</h3>
            <textarea
              rows="4"
              value={address}
              onChange={(e) => setAddress(e.target.value)}
              placeholder="Enter your shipping address..."
            />

            <h3>Payment Method</h3>
            <select
              value={paymentMethod}
              onChange={(e) => setPaymentMethod(e.target.value)}
            >
              <option value="COD">Cash on Delivery</option>
              <option value="UPI">UPI</option>
              <option value="Card">Credit/Debit Card</option>
            </select>

            {paymentMethod === 'UPI' && (
              <input
```

```
            type="text"
            placeholder="Enter your UPI ID"
            value={upiId}
            onChange={(e) => setUpiId(e.target.value)}
           />
          )}

          {paymentMethod === 'Card' && (
           <div className="card-inputs">
            <input
             type="text"
             placeholder="Card Number"
             maxLength={16}
             value={cardDetails.number}
             onChange={(e) => setCardDetails({ ...cardDetails, number:
e.target.value })}
            />
            <input
             type="text"
             placeholder="Valid Thru (MM/YY)"
             value={cardDetails.expiry}
             onChange={(e) => setCardDetails({ ...cardDetails, expiry:
e.target.value })}
            />
            <input
             type="password"
             placeholder="CVV"
             maxLength={3}
             value={cardDetails.cvv}
             onChange={(e) => setCardDetails({ ...cardDetails, cvv: e.target.value
})}
            />
           </div>
          )}

          <button onClick={handlePlaceOrder}>Place Order</button>
         </div>
        </>
      )}
    </div>
  );
}
```

```
export default Checkout;
```

CSS File - Cart.css

```
.cart-container {
  max-width: 1000px;
  margin: auto;
  padding: 30px 20px;
}

.cart-container h2 {
  text-align: center;
  margin-bottom: 20px;
}

.cart-item {
  display: flex;
  background: #fff;
  border-radius: 8px;
  margin-bottom: 20px;
  padding: 15px;
  box-shadow: 0 2px 5px rgba(0, 0, 0, 0.1);
}

.cart-item img {
  width: 140px;
  height: 140px;
  object-fit: cover;
  border-radius: 8px;
}

.details {
  flex: 1;
  margin-left: 20px;
}

.details h4 {
  margin: 0 0 10px;
}
```

```
.qty-controls {
  display: flex;
  align-items: center;
  gap: 10px;
  margin: 10px 0;
}

.qty-controls button {
  padding: 5px 10px;
  font-size: 16px;
  background: #eee;
  border: 1px solid #ccc;
  border-radius: 5px;
  cursor: pointer;
}

.remove-btn {
  background: #ff4d4d;
  color: white;
  border: none;
  padding: 6px 12px;
  border-radius: 4px;
  cursor: pointer;
}

.remove-btn:hover {
  background: #d93636;
}

.cart-total {
  text-align: right;
  margin-top: 30px;
}

.checkout-btn {
  padding: 10px 20px;
  background-color: #254dff;
  color: white;
  border: none;
  border-radius: 6px;
  font-size: 1rem;
  cursor: pointer;
```

```
}

.checkout-btn:hover {
  background-color: #1e40ff;
}

.empty-cart {
  text-align: center;
  margin-top: 50px;
}

.shop-now-btn {
  padding: 10px 20px;
  background-color: #ffcc00;
  color: black;
  border: none;
  border-radius: 6px;
  font-size: 1rem;
  cursor: pointer;
  margin-top: 20px;
}

.shop-now-btn:hover {
  background-color: #ffaa00;
}
```

Backend codes :

.env

```
MONGO_URI=mongodb://localhost:27017/toystore

JWT_SECRET=supersecretjwtkey
```

server.js

```
const express = require('express');
```

```
const mongoose = require('mongoose');
const cors = require('cors');
require('dotenv').config();

const productRoutes = require('./routes/products');
const authRoutes = require('./routes/auth');
const orderRoutes = require('./routes/orders');

const app = express();

app.use(cors());
app.use(express.json());

mongoose.connect(process.env.MONGO_URI, {
  useNewUrlParser: true,
  useUnifiedTopology: true,
}).then(() => console.log('MongoDB connected'))
  .catch((err) => console.error('MongoDB connection error:', err));

app.use('/api/products', productRoutes);
app.use('/api/auth', authRoutes);
app.use('/api/orders', orderRoutes);
```

```
const PORT = process.env.PORT || 5000;

app.listen(PORT, () => {

  console.log(`Server running on port ${PORT}`);

});
```

seed.js

```
const mongoose = require('mongoose');

const Product = require('./models/Product');

require('dotenv').config();


const sampleProducts = [
```

_id

:

ObjectId('6835d0da51efd4560aa9f076')

name

:

"Dancing Cactus"

image

:

"
EhUSEhI…"

price

:

600

__v

:

0

createdAt

:

2025-05-27T14:48:58.334+00:00

updatedAt

:

2025-05-27T14:48:58.334+00:00

_id

:

ObjectId('6835d0da51efd4560aa9f077')

name

:

"Toy Piano"

image

:

"
EhUTEhM…"

price

:

400

__v

:

0

createdAt

:

2025-05-27T14:48:58.334+00:00

updatedAt

:

2025-05-27T14:48:58.334+00:00

_id

:

ObjectId('6835d0da51efd4560aa9f078')

name

:

"Toy Light Saber"

image

:

"…"

price

:

700

__v

:

0

createdAt

:

2025-05-27T14:48:58.334+00:00

updatedAt

:

2025-05-27T14:48:58.334+00:00

_id

:

ObjectId('6835d0da51efd4560aa9f079')

name

:

"Laser Light"

image

:

"
hESERI…"

price

:

150

__v

:

0

createdAt

:

2025-05-27T14:48:58.334+00:00

updatedAt

:

2025-05-27T14:48:58.334+00:00

_id

:

ObjectId('6835d0da51efd4560aa9f07a')

name

:

"Toy Beyblade"

image

:

"https://www.jiomart.com/images/product/original/rvmzotkk7f/haridun-4-i…"

price

:

550

__v

:

0

createdAt

:

2025-05-27T14:48:58.334+00:00

updatedAt

:

2025-05-27T14:48:58.334+00:00

_id

:

ObjectId('6835d0da51efd4560aa9f07b')

name

:

"Barbie Phone"

image

:

"https://5.imimg.com/data5/SELLER/Default/2024/1/374807590/PS/GU/WS/205…"

price

:

120

__v

:

0

createdAt

:

2025-05-27T14:48:58.334+00:00

updatedAt

:

2025-05-27T14:48:58.334+00:00

];

```
mongoose.connect(process.env.MONGO_URI)
  .then(async () => {
    await Product.deleteMany({});
    await Product.insertMany(sampleProducts);
    console.log('Sample products inserted');
    process.exit();
  })
  .catch(err => console.error('Error seeding products:', err));
```

routes/auth.js

```
const express = require('express');
const jwt = require('jsonwebtoken');
const bcrypt = require('bcryptjs');
const User = require('../models/User');

const router = express.Router();

router.post('/register', async (req, res) => {
  try {
    const { name, email, password } = req.body;
    const existingUser = await User.findOne({ email });
    if (existingUser) return res.status(400).json({ error: 'Email already exists' });

    const user = new User({ name, email, password });
    await user.save();
    res.status(201).json({ message: 'User registered successfully' });
  } catch (err) {
    res.status(500).json({ error: 'Server error' });
  }
});
```

```
router.post('/login', async (req, res) => {
  try {
    const { email, password } = req.body;
    const user = await User.findOne({ email });
    if (!user || !(await bcrypt.compare(password, user.password))) {
      return res.status(400).json({ error: 'Invalid email or password' });
    }

    const token = jwt.sign({ userId: user._id }, process.env.JWT_SECRET, {
expiresIn: '7d' });
    res.json({ token, user: { name: user.name, email: user.email } });
  } catch (err) {
    res.status(500).json({ error: 'Server error' });
  }
});

module.exports = router;
```

routes/orders.js

```
const express = require('express');
const Order = require('../models/Order');
const jwt = require('jsonwebtoken');
const User = require('../models/User');
```

```
const router = express.Router();


function authMiddleware(req, res, next) {
  const authHeader = req.headers.authorization;
  if (!authHeader) return res.status(401).json({ error: 'No token provided' });


  const token = authHeader.split(' ')[1];
  try {
    const decoded = jwt.verify(token, process.env.JWT_SECRET);
    req.userId = decoded.userId;
    next();
  } catch {
    res.status(401).json({ error: 'Invalid token' });
  }
}


router.post('/', authMiddleware, async (req, res) => {
  try {
    const { items, total, shippingAddress } = req.body;
    const newOrder = new Order({ user: req.userId, items, total,
shippingAddress });
    await newOrder.save();
    res.status(201).json({ message: 'Order placed successfully', order: newOrder
});
```

```
  } catch (err) {
    res.status(500).json({ error: 'Failed to place order' });
  }
});

router.get('/', authMiddleware, async (req, res) => {
  try {
    const orders = await Order.find({ user: req.userId
}).populate('items.product');
    res.json(orders);
  } catch (err) {
    res.status(500).json({ error: 'Failed to fetch orders' });
  }
});

module.exports = router;
```

routes/products.js

```
const express = require('express');
const Product = require('../models/Product');
const router = express.Router();

router.get('/', async (req, res) => {
  try {
```

```
    const { search, category } = req.query;
    let filter = {};
    if (search) filter.name = { $regex: search, $options: 'i' };
    if (category) filter.category = category;

    const products = await Product.find(filter);
    res.json(products);
  } catch (err) {
    res.status(500).json({ error: 'Failed to fetch products' });
  }
});

router.get('/:id', async (req, res) => {
  try {
    const product = await Product.findById(req.params.id);
    if (!product) return res.status(404).json({ error: 'Product not found' });
    res.json(product);
  } catch (err) {
    res.status(500).json({ error: 'Failed to fetch product' });
  }
});

module.exports = router;
```

models/Order.js

```
const mongoose = require('mongoose');

const orderSchema = new mongoose.Schema({
  user: { type: mongoose.Schema.Types.ObjectId, ref: 'User', required: true },
  items: [
    {
      product: { type: mongoose.Schema.Types.ObjectId, ref: 'Product' },
      quantity: { type: Number, default: 1 },
    }
  ],
  total: { type: Number, required: true },
  shippingAddress: { type: String, required: true },
  createdAt: { type: Date, default: Date.now }
});

module.exports = mongoose.model('Order', orderSchema);
```

models/Product.js

```
const mongoose = require('mongoose');

const productSchema = new mongoose.Schema({
```

```
  name: { type: String, required: true },
  description: String,
  image: String,
  price: { type: Number, required: true },
  category: String,
}, { timestamps: true });

module.exports = mongoose.model('Product', productSchema);
```

models/User.js

```
const mongoose = require('mongoose');
const bcrypt = require('bcryptjs');

const userSchema = new mongoose.Schema({
  name: { type: String, required: true },
  email: { type: String, unique: true, required: true },
  password: { type: String, required: true }
});

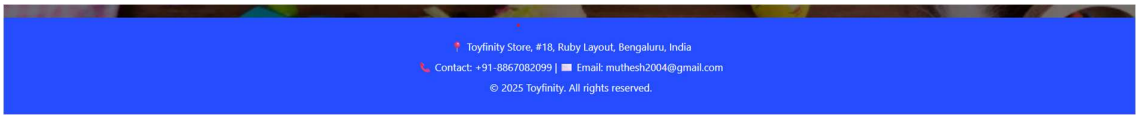
userSchema.pre('save', async function (next) {
  if (!this.isModified('password')) return next();
  try {
    this.password = await bcrypt.hash(this.password, 10);
```

```
    next();
  } catch (err) {
    next(err);
  }
});

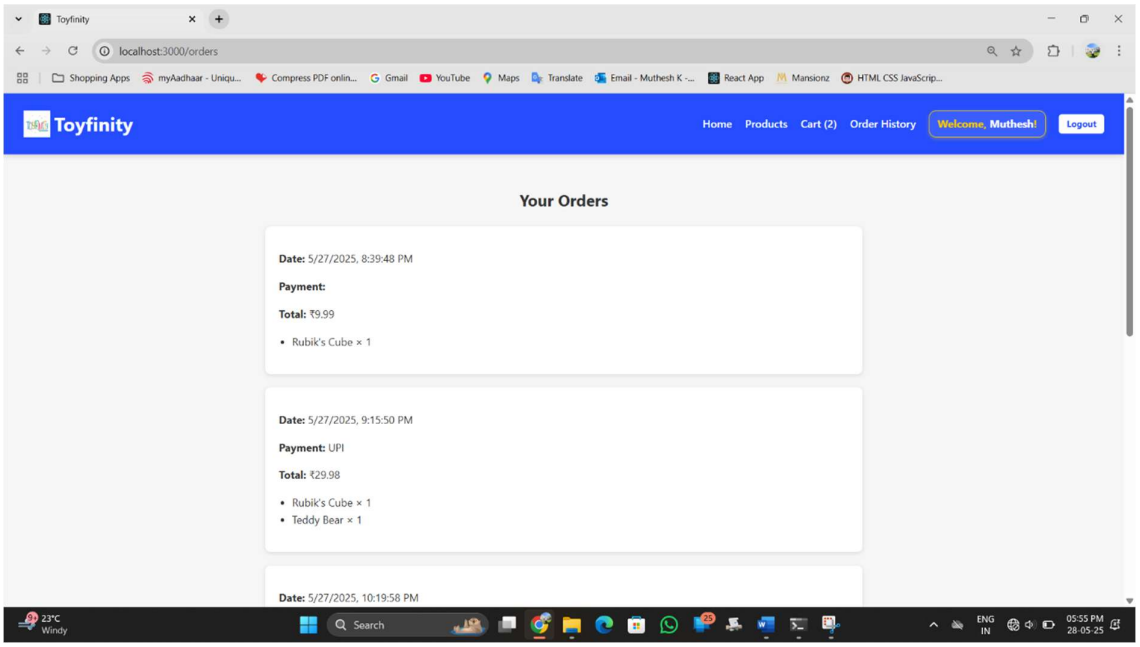
module.exports = mongoose.model('User', userSchema);
```

Project Visuals:-

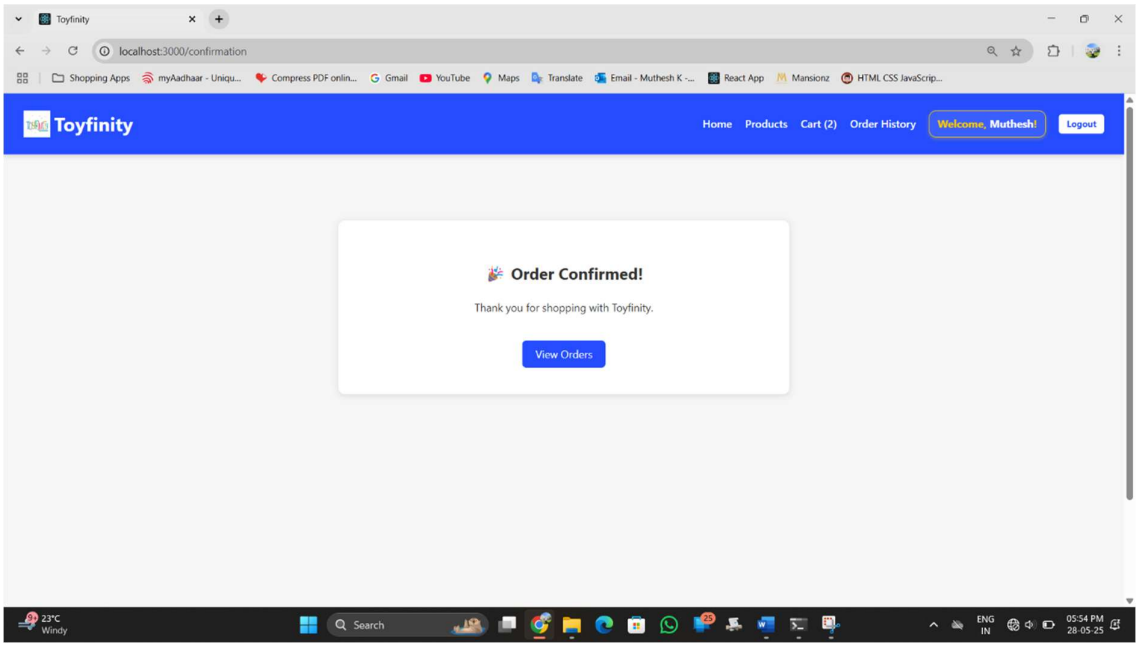
Footer:



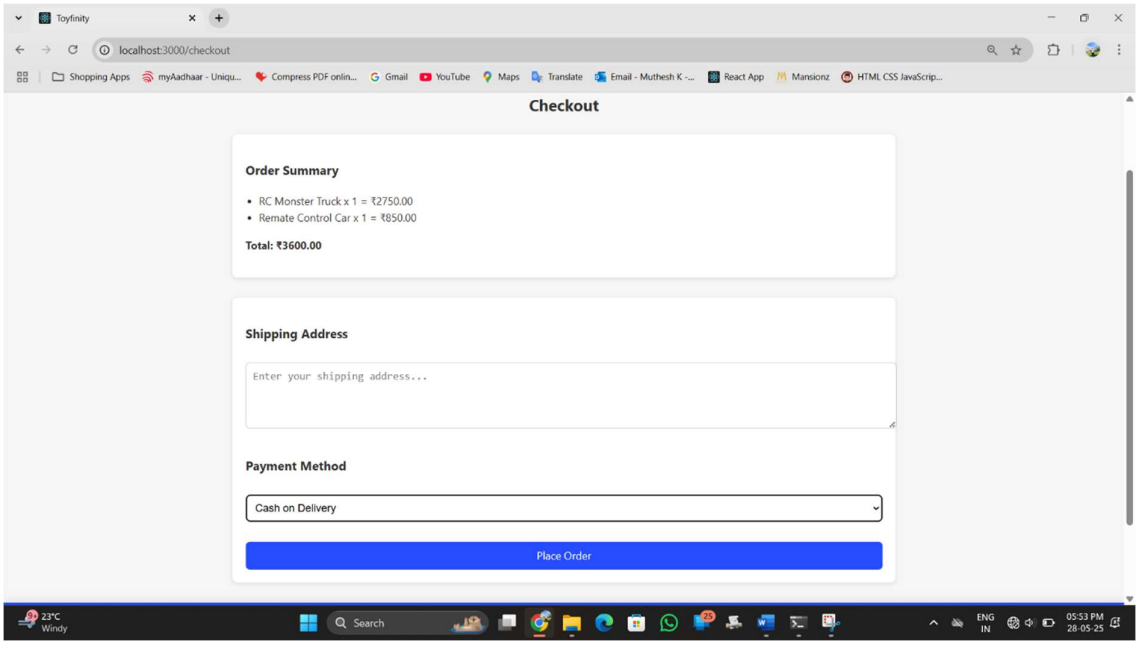
Orderhistory:



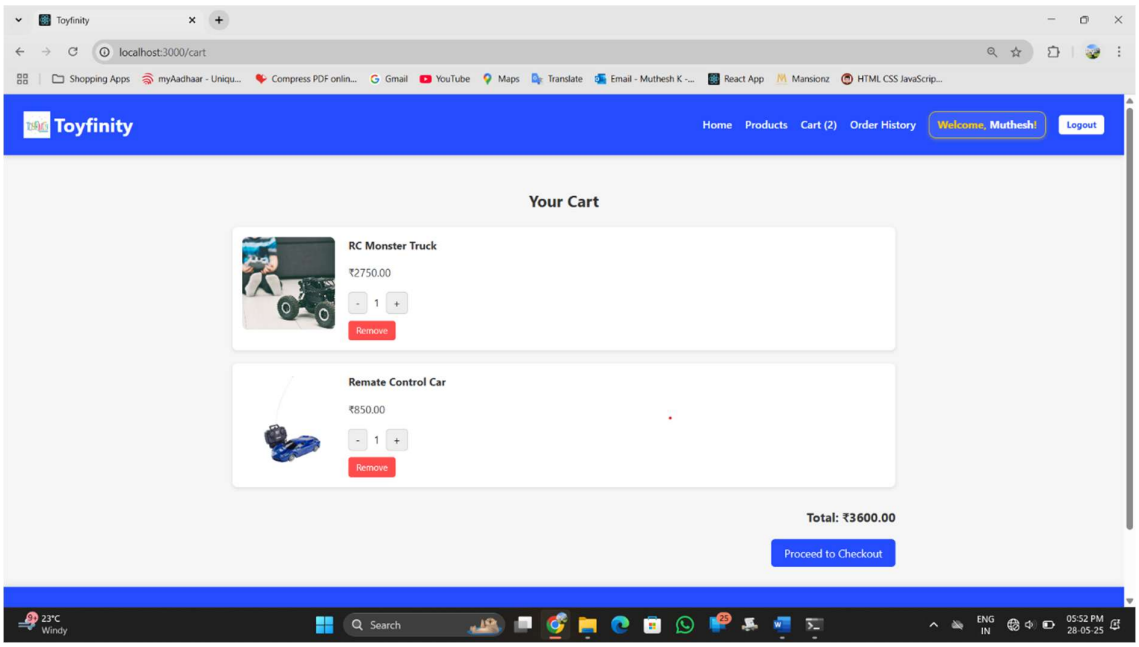
OrderConfirmation:



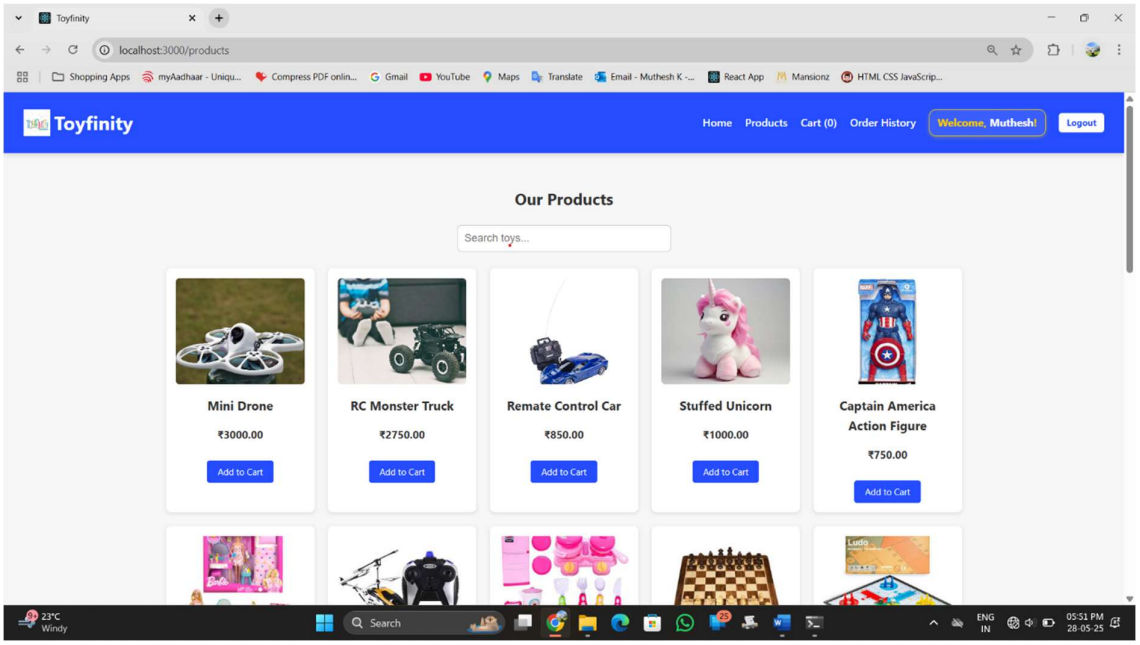
Checkout page:



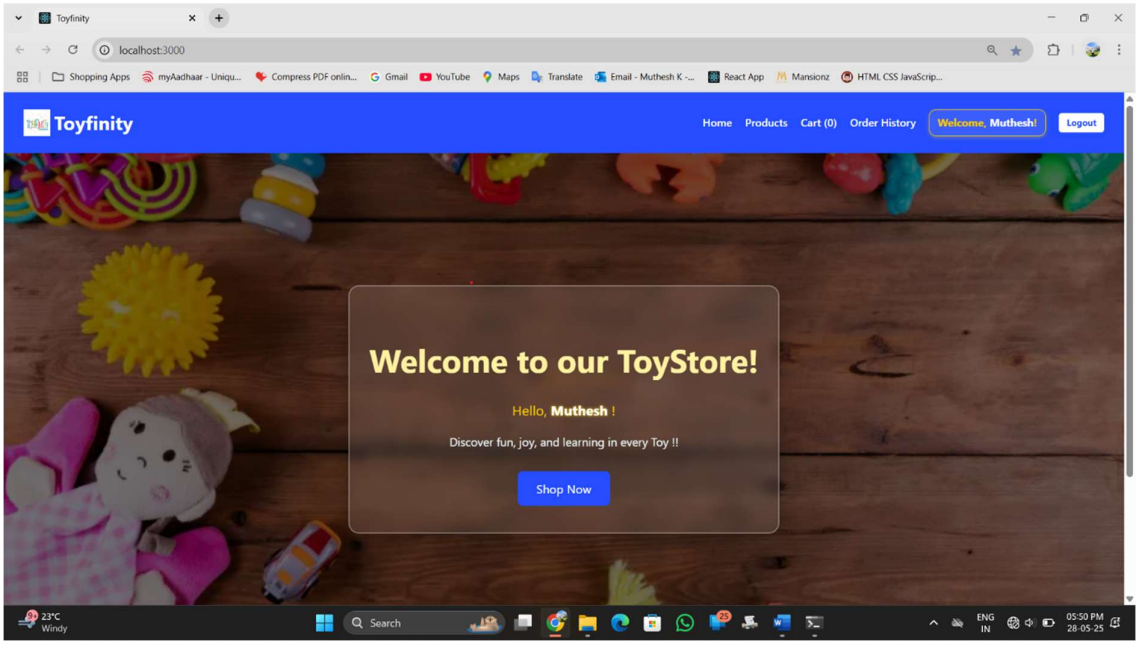
Cart Page:



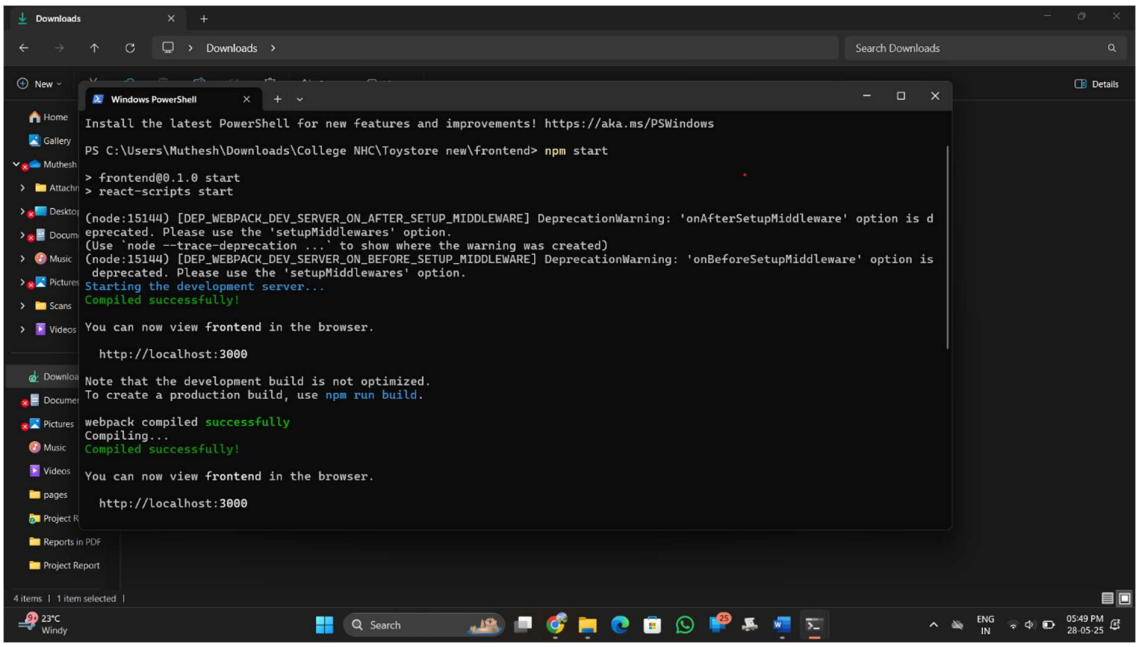
Products page:



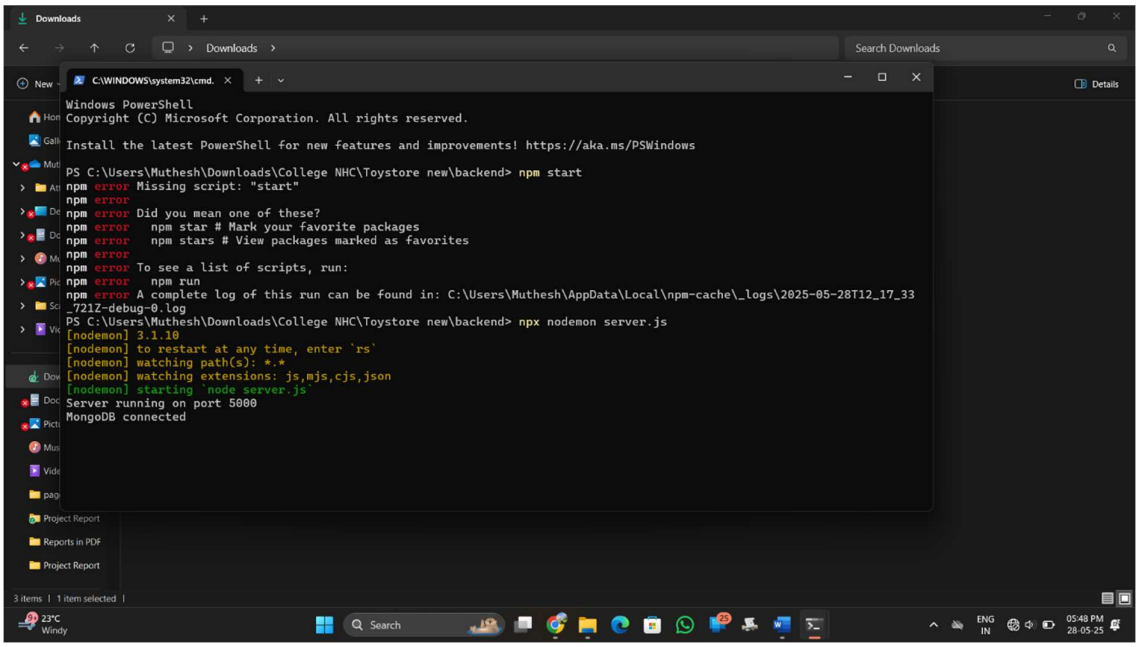
Home Page:



Frontend Starting:



Backend starting:



DECLARATION

I, Muthesh K (Reg. No. U19KU22S0111), a student of 6th Semester BCA, Department of Computer Applications, New Horizon College, hereby declare that the internship project work entitled “Toystore” has been carried out by me at Prinston Smart Engineers, Bengaluru, under the guidance of Mr. Abhishek (MERN Stack Developer). This report is submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Applications of Bangalore University during the academic year 2022–2023.

I also declare that, to the best of my knowledge and belief, the work presented in this report is original and has not formed the basis for the award of any degree, diploma, fellowship, or other similar titles by any other student.

Date: 01/05/2025
Place: Bengaluru

Muthesh K
U19KU22S0111

Bibliography(References):

1. **React Documentation**
   Meta Platforms, Inc. (2024). *React – A JavaScript library for building user interfaces*.
   Retrieved from: https://react.dev
2. **React Router Documentation**
   Remix Software Inc. (2024). *React Router – Declarative routing for React*.
   Retrieved from: https://reactrouter.com
3. **Node.js Documentation**
   OpenJS Foundation. (2024). *Node.js JavaScript runtime*.
   Retrieved from: https://nodejs.org
4. **Express.js Documentation**
   OpenJS Foundation. (2024). *Express – Fast, unopinionated, minimalist web framework for Node.js*.
   Retrieved from: https://expressjs.com
5. **MongoDB Documentation**
   MongoDB, Inc. (2024). *MongoDB – The developer data platform*.
   Retrieved from: https://www.mongodb.com/docs
6. **Mongoose Documentation**
   Automattic Inc. (2024). *Mongoose ODM – Elegant MongoDB object modeling for Node.js*.
   Retrieved from: https://mongoosejs.com
7. **Framer Motion Documentation**
   Framer Inc. (2024). *Framer Motion – Production-ready animation library for React*.
   Retrieved from: https://www.framer.com/motion/
8. **Font Awesome**
   Fonticons, Inc. (2024). *Font Awesome – Icons and social logos*.
   Retrieved from: https://fontawesome.com
9. **Dreamstime Image Source**
   Dreamstime. (n.d.). *Group of Baby Toys on Wooden Background* [Stock image].
   Retrieved from: https://www.dreamstime.com
10. **VS Code Editor**
    Microsoft Corporation. (2024). *Visual Studio Code – Code editing redefined*.
    Retrieved from: https://code.visualstudio.com